

Functions and Function Notation (Part II)

Function Notation Can Take Many Forms

① If $\Delta x = x^2 + 90$, find $\Delta 3$, $\Delta -2$, $\Delta 7$, and $\Delta -1$.

$$\Delta 3 = 3^2 + 90 = 9 + 90 = \boxed{99}$$

$$\Delta -2 = (-2)^2 + 90 = 4 + 90 = \boxed{94}$$

$$\Delta 7 = 7^2 + 90 = 49 + 90 = \boxed{139}$$

$$\Delta -1 = (-1)^2 + 90 = 1 + 90 = \boxed{91}$$

② If $A \odot B = A^3 + 2B$, find $-1 \odot 5$, $2 \odot 6$, $-4 \odot 1$, and $3 \odot 3$.

$$-1 \odot 5 = (-1)^3 + 2 \cdot 5 = -1 + 10 = \boxed{9}$$

$$2 \odot 6 = 2^3 + 2 \cdot 6 = 8 + 12 = \boxed{20}$$

$$-4 \odot 1 = (-4)^3 + 2 \cdot 1 = -64 + 2 = \boxed{-62}$$

$$3 \odot 3 = 3^3 + 2 \cdot 3 = 27 + 6 = \boxed{33}$$

③ If $\Delta y = \frac{4}{y}$, find $\Delta -2$, $\Delta 3$, and $\Delta -1$.

$$\Delta -2 = \frac{4}{-2} = \boxed{-2}$$

$$\Delta 3 = \frac{4}{3}$$

$$\Delta -1 = \frac{4}{-1} = \boxed{-4}$$

④ If $\Delta B = 2B - 5W$, find $\Delta -10$ and $\Delta 3$.

$$\Delta -10 = 2(-10) - 5(7) = -20 - 35 = \boxed{-55}$$

$$\Delta 3 = 2(3) - 5(4) = 6 - 20 = \boxed{-14}$$

Three Letters That Are Most Commonly Used for Function Notation

Letters	f	g	h
Examples	$f(x) = 2x - 1$	$g(x) = 5^{x+4}$	$h(x) = 3(x-6)^2$

⑤ If $f(x) = \frac{1}{8-2x}$, find $f(-6)$.

$$f(-6) = \frac{1}{8-2(-6)} = \frac{1}{8+12} = \frac{1}{20}$$

⑥ If $g(x) = 4^{x-1}$, find $g(-2)$.

$$g(-2) = 4^{-2-1} = 4^{-3} = \frac{1}{4^3} = \frac{1}{64}$$

⑦ If $h(x) = \frac{x^2}{3}$, find $h(1.2)$.

$$h(1.2) = \frac{1.2^2}{3} = \frac{1.44}{3} = \boxed{0.48}$$

⑧ If $f(x) = x + 7$, find $f(3)$.

$$f(3) = 3 + 7 = \boxed{10}$$

⑨ If $g(x) = \frac{3}{2}x + 10$, find $g(-4)$.

$$g(-4) = \frac{3}{2}(-4) + 10 = -6 + 10 = \boxed{4}$$

⑩ If $h(x) = \sqrt{5x}$, find $h(\frac{1}{5})$.

$$h(\frac{1}{5}) = \sqrt{5(\frac{1}{5})} = \sqrt{1} = \boxed{1}$$

Notation Indicating Operations between Functions

If "f" and "g" are functions, then

$$(f+g)(x) = f(x) + g(x)$$

$$(f-g)(x) = f(x) - g(x)$$

$$(f \cdot g)(x) = f(x) \cdot g(x)$$

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$$

$$(f \circ g)(x) = f(g(x))$$

⑪ If $f(x) = x^2 + 5x$ and $g(x) = 3x - 2$, find

$$A. (f+g)(x) = f(x) + g(x) = x^2 + 5x + 3x - 2 = \boxed{x^2 + 8x - 2}$$

$$\begin{aligned} B. (f-g)(x) &= f(x) - g(x) = x^2 + 5x - (3x - 2) \\ &= x^2 + 5x - 3x + 2 \\ &= \boxed{x^2 + 2x + 2} \end{aligned}$$

$$C. (f \cdot g)(x) = f(x) \cdot g(x) = \boxed{(x^2 + 5x)(3x - 2)}$$

$$D. \left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \boxed{\frac{x^2 + 5x}{3x - 2}}$$

E. $(f+g)(3)$

$$(f+g)(x) = x^2 + 8x - 2$$

$$(f+g)(3) = 3^2 + 8(3) - 2 = 9 + 24 - 2 = \boxed{31}$$

F. $(f-g)(-2)$

$$(f-g)(x) = x^2 + 2x + 2$$

$$(f-g)(-2) = (-2)^2 + 2(-2) + 2 = 4 - 4 + 2 = \boxed{2}$$

G. $(f \cdot g)(0)$

$$(f \cdot g)(x) = (x^2 + 5x)(3x - 2)$$

$$(f \cdot g)(0) = (0^2 + 5(0))(3(0) - 2) = (0 + 0)(0 - 2) = (0)(-2) = \boxed{0}$$

H. $\left(\frac{f}{g}\right)(10)$

$$\left(\frac{f}{g}\right)(x) = \frac{x^2 + 5x}{3x - 2}$$

$$\frac{f}{g}(10) = \frac{10^2 + 5(10)}{3(10) - 2} = \frac{100 + 50}{30 - 2} = \frac{150}{28} = \boxed{\frac{75}{14}}$$

12. If $f(x) = x + 1$ and $g(x) = 3^x - 6$, find

A. $(f+g)(x) = f(x) + g(x) = x + 1 + 3^x - 6 = \boxed{x + 3^x - 5}$

B. $(f-g)(x) = f(x) - g(x) = x + 1 - (3^x - 6) = x + 1 - 3^x + 6 = \boxed{x - 3^x + 7}$

C. $(f \cdot g)(x) = f(x)g(x) = \boxed{(x+1)(3^x-6)}$

D. $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \boxed{\frac{x+1}{3^x-6}}$

E. $(f+g)(2)$

$$(f+g)(x) = x + 3^x - 5$$

$$(f+g)(2) = 2 + 3^2 - 5 = 2 + 9 - 5 = \boxed{6}$$

F. $(f-g)(-1)$

$$(f-g)(x) = x - 3^x + 7$$

$$(f-g)(-1) = -1 - 3^{-1} + 7 = -1 - \frac{1}{3} + 7 = -1 - \frac{1}{3} + 7 = \boxed{5\frac{2}{3}}$$

G. $(f \cdot g)(1)$

$$(f \cdot g)(x) = (x+1)(3^x-6)$$

$$f \cdot g(1) = (1+1)(3^1-6) = (2)(3-6) = (2)(-3) = \boxed{-6}$$

H. $(\frac{f}{g})(0)$

$(\frac{f}{g})(x) = \frac{x+1}{3x-6}$

$(\frac{f}{g})(0) = \frac{0+1}{3(0)-6} = \frac{1}{-6} = \boxed{-\frac{1}{6}}$

⑬ If $f(x) = \frac{9}{6-x}$ and $g(x) = -2x$, find

A. $(f \circ g)(x) = f(g(x)) = \frac{9}{6-g(x)} = \frac{9}{6-(-2x)} = \boxed{\frac{9}{6+2x}}$

B. $(g \circ f)(x) = g(f(x)) = -2(f(x)) = -2(\frac{9}{6-x}) = \boxed{\frac{-18}{6-x}}$

C. $(f \circ g)(-2)$

$(f \circ g)(x) = \frac{9}{6+2x}$

$(f \circ g)(-2) = \frac{9}{6+2(-2)} = \frac{9}{6-4} = \boxed{\frac{9}{2}}$

D. $(g \circ f)(5)$

$(g \circ f)(x) = \frac{-18}{6-x} \quad \left\{ (g \circ f)(5) = \frac{-18}{6-5} = \boxed{-18} \right.$

⑭ If $f(x) = x^2 + 4x - 1$ and $g(x) = 2^x + 1$, find

A. $(f \circ g)(x) = f(g(x)) = (g(x))^2 + 4(g(x)) - 1 = \boxed{(2^x + 1)^2 + 4(2^x + 1) - 1}$

B. $(g \circ f)(x) = g(f(x)) = 2^{f(x)} + 1 = \boxed{2^{x^2 + 4x - 1} + 1}$

C. $(f \circ g)(1)$

$(f \circ g)(x) = (2^x + 1)^2 + 4(2^x + 1) - 1$

$(f \circ g)(1) = (2^1 + 1)^2 + 4(2^1 + 1) - 1 = (2+1)^2 + 4(2+1) - 1 = 3^2 + 4(3) - 1 = 9 + 12 - 1 = \boxed{20}$

D. $(g \circ f)(3)$

$(g \circ f)(x) = 2^{x^2 + 4x - 1} + 1$

$(g \circ f)(3) = 2^{3^2 + 4(3) - 1} + 1 = 2^{9+12-1} + 1 = \boxed{2^{20} + 1}$

⑮ If $f(x) = x+1$ and $g(x) = 6x$, find

A. $(f \circ g)(x) = f(g(x)) = g(x) + 1 = \boxed{6x+1}$

B. $(g \circ f)(x) = g(f(x)) = 6(f(x)) = 6(x+1) = \boxed{6x+6}$

C. $(f \circ g)(2)$

$$(f \circ g)(x) = 6x+1$$

$$(f \circ g)(2) = 6(2)+1 = 12+1 = \boxed{13}$$

D. $(g \circ f)(-3)$

$$(g \circ f)(x) = 6x+6$$

$$(g \circ f)(-3) = 6(-3)+6 = -18+6 = \boxed{-12}$$

⑯ If $f(x) = (x-2)^3 + x$ and $g(x) = 5x$, find

A. $(f \circ g)(x) = f(g(x)) = (g(x)-2)^3 + g(x)$
 $= \boxed{(5x-2)^3 + 5x}$

B. $(g \circ f)(x) = g(f(x)) = 5(f(x)) = 5[(x-2)^3 + x]$
 $= \boxed{5(x-2)^3 + 5x}$

C. $(f \circ g)(1)$

$$(f \circ g)(x) = (5x-2)^3 + 5x$$

$$(f \circ g)(1) = (5(1)-2)^3 + 5(1) = (5-2)^3 + 5$$
$$= 3^3 + 5$$
$$= \boxed{32}$$

D. $(g \circ f)(0)$

$$(g \circ f)(x) = 5(x-2)^3 + 5x$$

$$(g \circ f)(0) = 5(0-2)^3 + 5(0) = 5(-2)^3 + 0 = \boxed{-40}$$

Condensed Notation for Compound Inequalities

$a < x < b$ means that x is between a and b .

Example I: If $-3 < x < 5$, find three possible values of x .

$0, 1, 4$

Example II: If $8 \leq x < 100$, find three possible values of x .

$99, 50, 8$

Example III: If $4 \leq x \leq 6$, find three possible values of x .

$4, 6, 5.3$

Example IV: If $3 \leq x \leq 25$, find three possible values of x .

$7, 3, 25$

Example V: If $-18 < x < 16$, find three possible values of x .

$-15, 0, 15$

Example VI: If $-76 \leq x \leq -46$, find three possible values of x .

$-76, -46, -70$

Piecewise Functions

Piecewise Function: A function defined by multiple expressions, each with its own exclusive interval of inputs.

Example: $f(x) = \begin{cases} x-2, & 1 < x < 5 \\ 4+x^2, & 5 \leq x < 10 \end{cases}$

Given the functions below, find the values.

① $f(x) = \begin{cases} 2^x, & 0 < x < 4 \\ 5x-1, & 4 \leq x < 9 \end{cases}$

i) $f(3) = 2^3 = 8$

ii) $f(6) = 5(6)-1 = 30-1 = 29$

iii) $f(1) = 2^1 = 2$

iv) $f(5) = 5(5)-1 = 25-1 = 24$

v) $f(8) = 5(8)-1 = 40-1 = 39$

② $f(x) = \begin{cases} \frac{1}{x+2}, & -10 \leq x < -3 \\ x^3, & -3 \leq x \leq 8 \end{cases}$

i) $f(7) = 7^3 = 343$

ii) $f(0) = 0^3 = 0$

iii) $f(-4) = \frac{1}{-4+2} = \frac{1}{-2}$

iv) $f(-1) = (-1)^3 = -1$

$$(19) f(x) = \begin{cases} 3+x, & 2 < x < 7 \\ 4x, & 7 \leq x < 100 \end{cases}$$

$$i) f(9) = 4(9) = 36$$

$$ii) f(5) = 3+5 = 8$$

$$iii) f(30) = 4(30) = 120$$

$$iv) f(6.5) = 3+6.5 = 9.5$$

$$(20) f(x) = \begin{cases} \frac{1}{2}x, & -4 < x < 8 \\ \sqrt{x+7}, & 8 \leq x \leq 15 \end{cases}$$

$$i) f(0) = \frac{1}{2}(0) = 0$$

$$ii) f(9) = \sqrt{9+7} = \sqrt{16} = 4$$

$$iii) f(2) = \frac{1}{2}(2) = 1$$

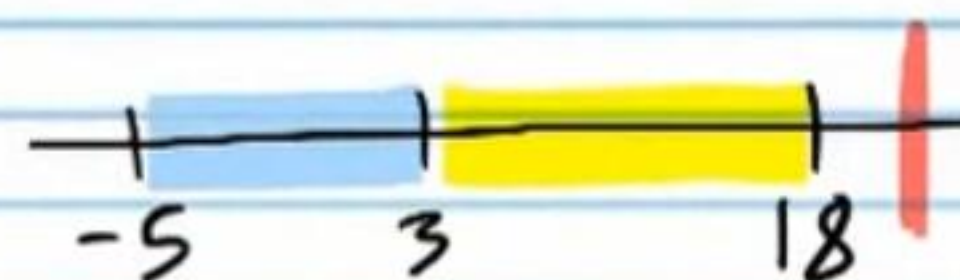
$$iv) f(14) = \sqrt{14+7} = \sqrt{21}$$

$$(21) f(x) = \begin{cases} -7(4^x)+1, & -5 \leq x < 3 \\ 4(2^x)-8, & 3 \leq x < 18 \end{cases}$$

$$i) f(3) = 4(2^3)-8 = 4(8)-8 = 32-8 = 24$$

$$ii) f(20) = \text{Undefined}$$

$$iii) f(-6) = \text{Undefined}$$



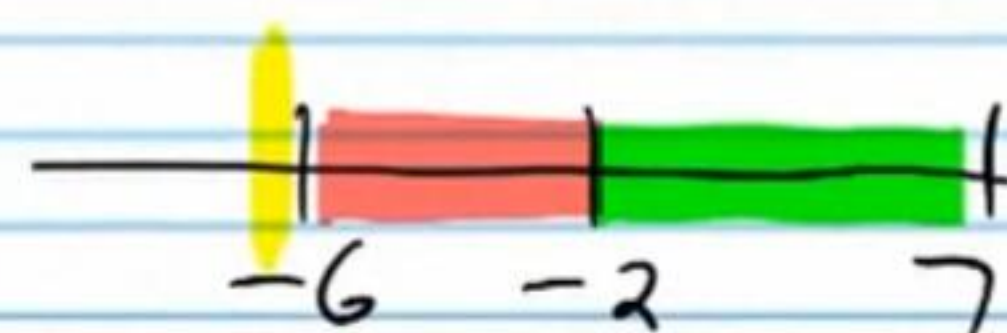
$$(22) f(x) = \begin{cases} 2, & -6 \leq x \leq -2 \\ 5+2x, & -2 < x \leq 7 \end{cases}$$

$$i) f(-2) = 2$$

$$ii) f(7) = 5+2(7) = 5+14 = 19$$

$$iii) f(-6) = 2$$

$$iv) f(-6.01) = \text{Undefined}$$



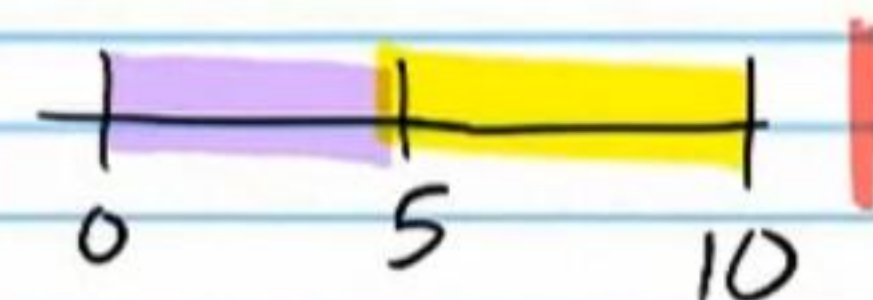
$$(23) f(x) = \begin{cases} 1+3^{x-2}, & 0 < x < 5 \\ x+4, & 5 \leq x \leq 10 \end{cases}$$

$$i) f(5) = 5+4 = 9$$

$$ii) f(10) = 10+4 = 14$$

$$iii) f(0) = \text{Undefined}$$

$$iv) f(12) = \text{Undefined}$$

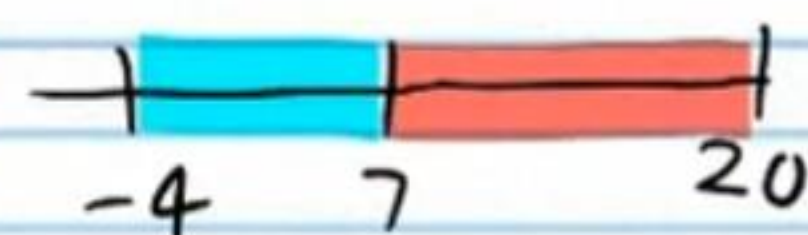


$$(24) f(x) = \begin{cases} x^2+5x+1, & -4 \leq x \leq 7 \\ 2x-3, & 7 < x < 20 \end{cases}$$

$$i) f(7) = 7^2+5(7)+1 = 49+35+1 = 85$$

$$ii) f(20) = \text{Undefined}$$

$$iii) f(-100) = \text{Undefined}$$



$$(25) f(x) = \begin{cases} 5x^2, & -10 < x < 4 \\ 1-7x, & 7 < x < 8 \end{cases}$$

$$i) f(5) = \text{Undefined}$$

$$ii) f(6) = \text{Undefined}$$

$$iii) f(0) = 5(0)^2 = 5(0) = 0$$

$$iv) f(7.4) = 1-7(7.4) = 1-51.8 = -50.8$$

$$v) f(4) = \text{Undefined}$$



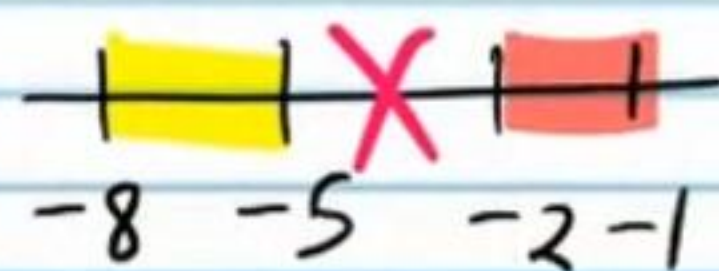
$$(26) f(x) = \begin{cases} x+2, & -8 \leq x \leq -5 \\ \frac{-3}{x}, & -2 \leq x \leq -1 \end{cases}$$

$$i) f(-4) = \text{Undefined}$$

$$ii) f(-1) = \frac{-3}{-1} = 3$$

$$iii) f(-2.7) = \text{Undefined}$$

$$iv) f(-5) = -5+2 = -3$$

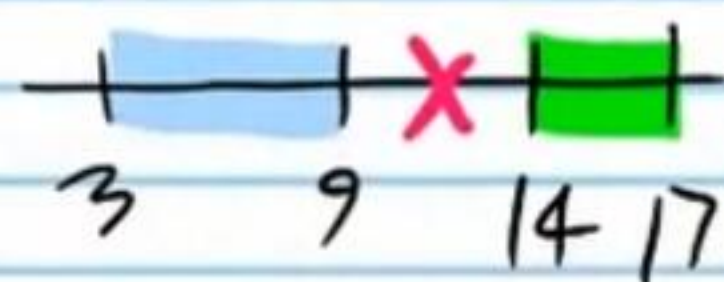


$$(27) f(x) = \begin{cases} \frac{1}{3}+x, & 3 < x < 9 \\ x+1, & 14 < x < 17 \end{cases}$$

$$i) f(14) = \text{Undefined}$$

$$ii) f(11) = \text{Undefined}$$

$$iii) f(9.1) = \text{Undefined}$$



$$(28) f(x) = \begin{cases} \frac{4}{x-7}, & -8 < x \leq -2 \\ \frac{2}{x+1}, & 2 < x \leq 4 \end{cases}$$

$$i) f(-8) = \text{Undefined}$$

$$ii) f(-2) = \frac{4}{-2-7} = \frac{4}{-9}$$

$$iii) f(0) = \text{Undefined}$$

$$iv) f(1) = \text{Undefined}$$

